

Lecture 2

Quantum States

Art: *Oddly enough, that beer made my head* **stop** *spinning. What state are we in?*

Lenny: *I wish I knew. Does it matter?*

Art: *It might. I don't think we're in California anymore*.

2.1 STATES AND VECTORS

In classical physics, knowing *the state of a system* implies knowing everything that is necessary to predict the future of that system. As we've seen in the last lecture, quantum systems are not completely predictable. Evidently, quantum states have a different meaning than classical states. Very roughly, knowing a quantum state means knowing *as much as can be known* about how the system was prepared. In the last chapter, we talked about using an apparatus to prepare the state of a spin. In fact, we implicitly assumed that there was no more fine detail to specify or that could be specified about the state of the spin.

The obvious question to ask is whether the unpredictability is due to an incompleteness in what we call a quantum state. There are various opinions about this matter. Here is a sampling: